1. Write a Relational Algebra expression to perform **Cartesian Product (Cross Join)** between Emp and Dept tables.

➔ select e.emp_id, e.emp_name, d.dep_name, d.location from employee as e

cross join department as d ;

emp_id	emp_name	dep_name	location
101	vishva	testing	banglore
101	vishva	finanace	nashik
101	vishva	development	pune
101	vishva	HR	mumbai
101	vishva	IT	mumbai
101	vishva	Computer	pune
102	sanika	testing	banglore
102	sanika	finanace	nashik
102	sanika	development	pune
102	sanika	HR	mumbai
102	sanika	IT	mumbai
102	sanika	Computer	pune
103	sakshi	testing	banglore
103	sakshi	finanace	nashik
103	sakshi	development	pune
103	sakshi	HR	mumbai
103	sakshi	IT	mumbai
103	sakshi	Computer	pune
104	vaishali	testing	banglore
104	vaishali	finanace	nashik
104	vaishali	development	pune
104	vaishali	HR	mumbai
104	vaishali	IT	mumbai
104	vaishali	Computer	pune
105	Sneha	testing	banglore
105	Sneha	finanace	nashik
105	Sneha	development	pune
105	Sneha	HR	mumbai
105	Sneha	IT	mumbai
105	Sneha	Computer	pune
106	Neha	testing	banglore
106	Neha	finanace	nashik
106	Neha	development	pune
106	Neha	HR	mumbai
106	Neha	IT	mumbai
106	Neha	Computer	pune
107	Karan	testing	banglore
107	Karan	finanace	nashik
107	Karan	development	pune
107	Karan	HR	mumbai
107	Karan	IT	mumbai
107	Karan	Computer	pune
108	Pooja	testing	banglore
108	Pooja	finanace	nashik
108	Pooja	development	pune
108	Pooja	HR	mumbai
108	Pooja	IT	mumbai
108	Pooja	Computer	pune
109	Anjali	testing	banglore
109	Anjali	finanace	nashik
109	Anjali	development	pune
109	Anjali	HR	mumbai
109	Anjali	IT	mumbai
109	Anjali	Computer	pune
110	Priya	testing	banglore
110	Priya	finanace	nashik
110	Priya	development	pune
110	Priya	HR	mumbai
110	Priya	IT	mumbai
110	Priya	Computer	pune

Q2 . Write a SQL query to perform an **INNER JOIN** between Emp and Dept tables displaying EmpName and DeptName.

➔ select e.emp_name,d.dep_name from employee as e
inner join department as d on e.dep_id = d.dep_id;

```
mysql> select e.emp_name,d.dep_name from employee as e
    -> inner join department as d on e.dep_id = d.dep_id;
+----------+-------------+
| emp_name | dep_name    |
+----------+-------------+
| vishva   | Computer    |
| Anjali   | Computer    |
| sakshi   | IT          |
| Pooja    | IT          |
| vaishali | HR          |
| Sneha    | HR          |
| Neha     | development |
| Priya    | development |
| Karan    | finanace    |
| sanika   | testing     |
+----------+-------------+
10 rows in set (0.00 sec)
```

Q3. Write a SQL query to perform a **LEFT OUTER JOIN** between Emp and Dept tables to display all employees along with their department information, including employees without a department.

➔ select * from employee as e left outer join department as d on e.dep_id = d.dep_id;

```
mysql> select * from employee as e left outer join department as d on e.dep_id = d.dep_id;
+--------+----------+----------+--------+------------+--------+-------------+----------+
| emp_id | emp_name | salary   | dep_id | project_id | dep_id | dep_name    | location |
+--------+----------+----------+--------+------------+--------+-------------+----------+
|    101 | vishva   | 85000.00 |     10 |          1 |     10 | Computer    | pune     |
|    102 | sanika   | 45000.00 |     60 |          2 |     60 | testing     | banglore |
|    103 | sakshi   | 50000.00 |     20 |          4 |     20 | IT          | mumbai   |
|    104 | vaishali | 60000.00 |     30 |          3 |     30 | HR          | mumbai   |
|    105 | Sneha    | 90000.00 |     30 |          4 |     30 | HR          | mumbai   |
|    106 | Neha     | 55000.00 |     40 |          5 |     40 | development | pune     |
|    107 | Karan    | 80000.00 |     50 |          1 |     50 | finanace    | nashik   |
|    108 | Pooja    | 40000.00 |     20 |          3 |     20 | IT          | mumbai   |
|    109 | Anjali   | 65000.00 |     10 |          2 |     10 | Computer    | pune     |
|    110 | Priya    | 75000.00 |     40 |          5 |     40 | development | pune     |
+--------+----------+----------+--------+------------+--------+-------------+----------+
10 rows in set (0.00 sec)
```

Q4. Write a SQL query to perform a **RIGHT OUTER JOIN** between Emp and Dept tables

➔ select * from employee as e right outer join department as d on e.dep_id = d.dep_id;

```
mysql> select * from employee as e right outer join department as d on e.dep_id = d.dep_id;
+--------+----------+----------+--------+------------+--------+-------------+----------+
| emp_id | emp_name | salary   | dep_id | project_id | dep_id | dep_name    | location |
+--------+----------+----------+--------+------------+--------+-------------+----------+
|    101 | vishva   | 85000.00 |     10 |          1 |     10 | Computer    | pune     |
|    109 | Anjali   | 65000.00 |     10 |          2 |     10 | Computer    | pune     |
|    103 | sakshi   | 50000.00 |     20 |          4 |     20 | IT          | mumbai   |
|    108 | Pooja    | 40000.00 |     20 |          3 |     20 | IT          | mumbai   |
|    104 | vaishali | 60000.00 |     30 |          3 |     30 | HR          | mumbai   |
|    105 | Sneha    | 90000.00 |     30 |          4 |     30 | HR          | mumbai   |
|    106 | Neha     | 55000.00 |     40 |          5 |     40 | development | pune     |
|    110 | Priya    | 75000.00 |     40 |          5 |     40 | development | pune     |
|    107 | Karan    | 80000.00 |     50 |          1 |     50 | finanace    | nashik   |
|    102 | sanika   | 45000.00 |     60 |          2 |     60 | testing     | banglore |
+--------+----------+----------+--------+------------+--------+-------------+----------+
10 rows in set (0.00 sec)
```

Q5. Write a SQL query to perform a **NATURAL JOIN** between Emp and Dept.

➔ select * from employee natural join department ;

```
mysql> select * from employee natural join department ;
+--------+--------+----------+----------+------------+-------------+----------+
| dep_id | emp_id | emp_name | salary   | project_id | dep_name    | location |
+--------+--------+----------+----------+------------+-------------+----------+
|     10 |    101 | vishva   | 85000.00 |          1 | Computer    | pune     |
|     10 |    109 | Anjali   | 65000.00 |          2 | Computer    | pune     |
|     20 |    103 | sakshi   | 50000.00 |          4 | IT          | mumbai   |
|     20 |    108 | Pooja    | 40000.00 |          3 | IT          | mumbai   |
|     30 |    104 | vaishali | 60000.00 |          3 | HR          | mumbai   |
|     30 |    105 | Sneha    | 90000.00 |          4 | HR          | mumbai   |
|     40 |    106 | Neha     | 55000.00 |          5 | development | pune     |
|     40 |    110 | Priya    | 75000.00 |          5 | development | pune     |
|     50 |    107 | Karan    | 80000.00 |          1 | finanace    | nashik   |
|     60 |    102 | sanika   | 45000.00 |          2 | testing     | banglore |
+--------+--------+----------+----------+------------+-------------+----------+
10 rows in set (0.00 sec)
```

Q6. Write a SQL query to perform a **CROSS JOIN** between Emp and Project table.

➔ select e.emp_name,p.project_name from employee as e cross join project as p;

```
mysql> select e.emp_name,p.project_name from employee as e
    -> cross join project as p;
+----------+--------------------+
| emp_name | project_name       |
+----------+--------------------+
| vishva   | Cloud Migration    |
| vishva   | Data Analytics     |
| vishva   | Payroll System     |
| vishva   | Mobile Application |
| vishva   | Website Development |
| sanika   | Cloud Migration    |
| sanika   | Data Analytics     |
| sanika   | Payroll System     |
| sanika   | Mobile Application |
| sanika   | Website Development |
| sakshi   | Cloud Migration    |
| sakshi   | Data Analytics     |
| sakshi   | Payroll System     |
| sakshi   | Mobile Application |
| sakshi   | Website Development |
| vaishali | Cloud Migration    |
| vaishali | Data Analytics     |
| vaishali | Payroll System     |
| vaishali | Mobile Application |
| vaishali | Website Development |
| Sneha    | Cloud Migration    |
| Sneha    | Data Analytics     |
| Sneha    | Payroll System     |
| Sneha    | Mobile Application |
| Sneha    | Website Development |
| Neha     | Cloud Migration    |
| Neha     | Data Analytics     |
| Neha     | Payroll System     |
| Neha     | Mobile Application |
| Neha     | Website Development |
| Karan    | Cloud Migration    |
| Karan    | Data Analytics     |
| Karan    | Payroll System     |
| Karan    | Mobile Application |
| Karan    | Website Development |
| Pooja    | Cloud Migration    |
| Pooja    | Data Analytics     |
| Pooja    | Payroll System     |
| Pooja    | Mobile Application |
| Pooja    | Website Development |
| Anjali   | Cloud Migration    |
| Anjali   | Data Analytics     |
| Anjali   | Payroll System     |
| Anjali   | Mobile Application |
| Anjali   | Website Development |
| Priya    | Cloud Migration    |
| Priya    | Data Analytics     |
| Priya    | Payroll System     |
| Priya    | Mobile Application |
| Priya    | Website Development |
+----------+--------------------+
50 rows in set (0.00 sec)
```

Q7. Write a SQL query to **create a new table EmpBackup with the same structure as Emp but no data**.

➔ create table emp_backup(emp_id int primary key, emp_name varchar(30), salary decimal(10,2), dep_id int, project_id int , foreign key (dep_id) references department(dep_id));

```
mysql> desc emp_backup;
+------------+--------------+------+-----+---------+-------+
| Field      | Type         | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
| emp_id     | int          | NO   | PRI | NULL    |       |
| emp_name   | varchar(30)  | YES  |     | NULL    |       |
| salary     | decimal(10,2)| YES  |     | NULL    |       |
| dep_id     | int          | YES  | MUL | NULL    |       |
| project_id | int          | YES  |     | NULL    |       |
+------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

Q8. Write a SQL query to **copy all data from Emp into EmpBackup table**.

➔ insert into emp_backup select emp_id, emp_name, salary,dep_id, project_id from employee;

```
mysql> select * from emp_backup;
+--------+----------+----------+--------+------------+
| emp_id | emp_name | salary   | dep_id | project_id |
+--------+----------+----------+--------+------------+
|    101 | vishva   | 85000.00 |     10 |          1 |
|    102 | sanika   | 45000.00 |     60 |          2 |
|    103 | sakshi   | 50000.00 |     20 |          4 |
|    104 | vaishali | 60000.00 |     30 |          3 |
|    105 | Sneha    | 90000.00 |     30 |          4 |
|    106 | Neha     | 55000.00 |     40 |          5 |
|    107 | Karan    | 80000.00 |     50 |          1 |
|    108 | Pooja    | 40000.00 |     20 |          3 |
|    109 | Anjali   | 65000.00 |     10 |          2 |
|    110 | Priya    | 75000.00 |     40 |          5 |
+--------+----------+----------+--------+------------+
10 rows in set (0.00 sec)
```

Q9. Write a SQL query to **create a new table ProjectArchive with the same structure and data as Project**.

➔ create table ProjectArchive(project_id int primary key, project_name varchar(20), budget decimal(10,2));

insert into ProjectArchive select project_id, project_name, budget from project;

```
mysql> select * from ProjectArchive;
+------------+--------------------+-----------+
| project_id | project_name       | budget    |
+------------+--------------------+-----------+
|          1 | Website Development | 200000.00 |
|          2 | Mobile Application  | 300000.00 |
|          3 | Payroll System      | 150000.00 |
|          4 | Data Analytics      | 250000.00 |
|          5 | Cloud Migration     | 400000.00 |
+------------+--------------------+-----------+
5 rows in set (0.00 sec)
```

Q10. Write a SQL query to create an **AUTO_INCREMENT sequence for EmpID in the Emp table** during table creation.

➔ alter table employee modify emp_id int not null auto_increment;

```
mysql> desc employee;
+------------+--------------+------+-----+---------+----------------+
| Field      | Type         | Null | Key | Default | Extra          |
+------------+--------------+------+-----+---------+----------------+
| emp_id     | int          | NO   | PRI | NULL    | auto_increment |
| emp_name   | varchar(30)  | YES  |     | NULL    |                |
| salary     | decimal(10,2)| YES  |     | NULL    |                |
| dep_id     | int          | YES  | MUL | NULL    |                |
| project_id | int          | YES  |     | NULL    |                |
+------------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

Q11. Write a SQL query using a **subquery** to find all employees whose salary is greater than the **average salary of all employees**.

➔ select emp_id,emp_name, salary from employee where salary > (select avg(salary) from employee);

```
mysql> select emp_id,emp_name, salary from employee where salary > (select avg(salary) from employee);
+--------+----------+----------+
| emp_id | emp_name | salary   |
+--------+----------+----------+
|    101 | vishva   | 85000.00 |
|    105 | Sneha    | 90000.00 |
|    107 | Karan    | 80000.00 |
|    109 | Anjali   | 65000.00 |
|    110 | Priya    | 75000.00 |
+--------+----------+----------+
5 rows in set (0.00 sec)
```

Q12. Write a SQL query using a **subquery in WHERE clause** to find employees working in departments located in 'Mumbai'

➔ select emp_id,emp_name, salary, dep_id from employee where dep_id in (select dep_id from department where location = 'mumbai');

```
mysql> select emp_id,emp_name, salary, dep_id from employee where dep_id in (select dep_id from department where location = 'mumbai');
+--------+----------+----------+--------+
| emp_id | emp_name | salary   | dep_id |
+--------+----------+----------+--------+
|    103 | sakshi   | 50000.00 |     20 |
|    108 | Pooja    | 40000.00 |     20 |
|    104 | vaishali | 60000.00 |     30 |
|    105 | Sneha    | 90000.00 |     30 |
+--------+----------+----------+--------+
4 rows in set (0.00 sec)
```

Q13. Write a SQL query to display all departments where the number of employees is greater than **5**, using a subquery.

➔

```
mysql> select dep_id, dep_name from department
    -> where dep_id in(
    -> select dep_id from employee
    -> group by dep_id
    -> having count(*) >5);
Empty set (0.07 sec)
```

Q14. Write a subqueries for find out Max salary of employee.

➔ select emp_id, emp_name , salary from employee where salary = (select max(salary) from employee);

```
mysql> select emp_id, emp_name , salary from employee where salary = (select max(salary) from employee);
+--------+----------+----------+
| emp_id | emp_name | salary   |
+--------+----------+----------+
|    105 | Sneha    | 90000.00 |
+--------+----------+----------+
1 row in set (0.00 sec)
```

Q15. Write a self join query for each table.

➔ select e1.emp_name as empName1 ,e2.emp_name as empName2 , e1.dep_id from employee as e1 join employee as e2 where e1.dep_id = e2.dep_id;

```
mysql> select e1.emp_name as empName1 ,e2.emp_name as empName2 , e1.dep_id from employee as e1
    -> join employee as e2 on e1.dep_id = e2.dep_id;
+----------+----------+--------+
| empName1 | empName2 | dep_id |
+----------+----------+--------+
| vishva   | vishva   |     10 |
| vishva   | Anjali   |     10 |
| sanika   | sanika   |     60 |
| sakshi   | sakshi   |     20 |
| sakshi   | Pooja    |     20 |
| vaishali | vaishali |     30 |
| vaishali | Sneha    |     30 |
| Sneha    | vaishali |     30 |
| Sneha    | Sneha    |     30 |
| Neha     | Neha     |     40 |
| Neha     | Priya    |     40 |
| Karan    | Karan    |     50 |
| Pooja    | sakshi   |     20 |
| Pooja    | Pooja    |     20 |
| Anjali   | vishva   |     10 |
| Anjali   | Anjali   |     10 |
| Priya    | Neha     |     40 |
| Priya    | Priya    |     40 |
+----------+----------+--------+
18 rows in set (0.00 sec)
```

select d1.dep_name as depName1 ,d2.dep_name as depName2 , d1.location from department as d1 join department as d2 on d1.location = d2.location ;

```
mysql> select d1.dep_name as depName1 ,d2.dep_name as depName2 , d1.location from department as d1
    -> join department as d2 on d1.location = d2.location ;
+-------------+-------------+----------+
| depName1    | depName2    | location |
+-------------+-------------+----------+
| development | Computer    | pune     |
| Computer    | Computer    | pune     |
| HR          | IT          | mumbai   |
| IT          | IT          | mumbai   |
| HR          | HR          | mumbai   |
| IT          | HR          | mumbai   |
| development | development | pune     |
| Computer    | development | pune     |
| finanace    | finanace    | nashik   |
| testing     | testing     | banglore |
+-------------+-------------+----------+
10 rows in set (0.00 sec)
```

select p1.project_name as projectName1 ,p2.project_name as projectName2 , p1.budget from project as p1 join project as p2 on p1.budget = p2.budget;

```
mysql> select p1.project_name as projectName1 ,p2.project_name as projectName2 , p1.budget from project as p1
    -> join project as p2 on p1.budget = p2.budget;
+--------------------+--------------------+-----------+
| projectName1       | projectName2       | budget    |
+--------------------+--------------------+-----------+
| Website Development | Website Development | 200000.00 |
| Mobile Application | Mobile Application | 300000.00 |
| Payroll System     | Payroll System     | 150000.00 |
| Data Analytics     | Data Analytics     | 250000.00 |
| Cloud Migration    | Cloud Migration    | 400000.00 |
+--------------------+--------------------+-----------+
5 rows in set (0.07 sec)
```